KISHORE KUMAR SUBRAMANIAN

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ASPIRING VLSI ENGINEER | RTL DESIGN, VERIFICATION & PHYSICAL DESIGN

SUMMARY

Electronics and Communication Engineering student with hands-on experience in **digital system design and FPGA development** using **Verilog and Vivado**. Worked with **Zynq-7000 FPGA platforms** and embedded hardware through practical projects with documented workflows on GitHub, and implemented the **RTL-to-GDS physical design flow** using open-source EDA tools on the **Sky130 PDK**. Interested in gaining research experience in **RTL design, digital systems, and VLSI physical design**.

TECHNICAL SKILLS

HDL & RTL: Verilog, RTL Design, Digital System Design
VLSI: Physical Design Flow, RTL-to-GDS, Synthesis, Floorplanning, Placement, CTS, Routing, DRC, LVS
FPGA Tools: Xilinx Vivado, Zedboard, Zynq-7000
EDA Tools: Yosys, OpenROAD, Magic, Netgen
Digital Systems: Combinational Circuits, Sequential Circuits, FSM Design
Programming: Python

PROJECTS

Synchronous Bidirectional Counter (Sky130 PDK) – RTL-to-GDS Flow 2026

- Designed and implemented a complete **RTL-to-GDS physical design flow** for an **8-bit synchronous up/down counter** using open-source EDA tools.
- Performed synthesis using **Yosys**, followed by floorplanning, placement, clock tree synthesis, and routing using **OpenROAD**.
- Generated layout and GDS using **Magic** and performed **LVS verification using Netgen** on the SkyWater **Sky130 PDK**.

Zynq-7000 GPIO Logic Workbench – github.com/kishlabs/zynq7000-gpio-logic-workbench 2026

- Developed GPIO-based digital logic experiments using **Zynq-7000 FPGA** on Zedboard.
- Implemented hardware interaction using board switches and LEDs.
- Used **Vivado block design** for FPGA configuration and bitstream generation.
- Documented a reproducible FPGA workflow for experimentation.

CERTIFICATIONS

- **VLSI Physical Design and Timing Analysis** – NPTEL (Elite + Silver), May 2026
- **VLSI Design Flow: RTL to GDS** – NPTEL (Elite + Silver), Nov 2025
- **Digital System Design** – NPTEL (Elite + Silver), May 2025
- **Learning FPGA Development; Learning Verilog for FPGA Development** – LinkedIn Learning, June 2025
- **VLSI for Beginners** – NIELT, Feb 2025

AWARDS & ACHIEVEMENTS

- **Runner-up**, InnovateX Hackathon, Presidency University Bengaluru (2025) – Rs. 15,000 for a V2V prototype built in 24 hours
- **Best Innovation Award**, HackTronix Hardware Hackathon, Chennai (2025) – Rs. 2,000 for a hardware prototype
- **Top 70 of 2500+**, SPRINTATHON'25 National Hackathon, Chennai (2025) – Carmunicate, a real-time V2V communication device
- **Shortlisted** in multiple national-level hackathons

EDUCATION

B.E. Electronics and Communication Engineering 2023 – 2027
Government College of Engineering, Erode Tamil Nadu

- **CGPA**: 8.01 (up to 6th Semester)
- **Relevant Coursework**: Digital Electronics, Digital System Design, VLSI, Embedded Systems

Higher Secondary) 2022
IIP Laxmi Raman School Tirunelveli, Tamil Nadu

- **Percentage**: 90.1%
- **Relevant Coursework**: Maths & Computer Science

TECHNICAL INTERESTS

RTL Design | FPGA Development | Digital System Design | VLSI Physical Design

LANGUAGES KNOWN

Tamil – Native | **English** – Full professional | **Hindi** – Reading and writing